

Python Exercise for Lecture 7

Law of Large Numbers (LLN) and Central Limit Theorem (CLT)

We simulate i.i.d. random variables

$$X \sim U(a, b) = U(0, 10)$$

Import packages

```
import numpy as np
import matplotlib.pyplot as plt
```

Part 1 — Exploring a single sample

Let the sample size be

$$n = 100.$$

Generate one sample:

```
a = 0
b = 10
n = 100

x = np.random.uniform(a, b, n)
```

1. Plot a histogram of the sample

```
plt.hist(x, bins=10, edgecolor="black")
```

2. Change the number of bins: 5, 10, 100

Questions

- How does the appearance change when you increase the number of bins?
- Which bin choice seems most informative for $n = 100$?

3. Compute the sample mean and sample variance

Part 2 — Law of Large Numbers (LLN)

Now investigate how the sample mean behaves as the sample size increases.

Use sample sizes

$$n = 10, 100, 1000.$$

For each n , generate a new sample and compute the sample mean:

Part 3 — Repeated experiments (distribution of the sample mean)

Now fix the sample size

$$n = 30$$

and repeat the experiment many times:

$$R = 1000.$$

```
a = 0
b = 10
n = 30
R = 1000

means = [] # empty list to store sample means

for i in range(R):
    x = np.random.uniform(a, b, n)
    means.append(np.mean(x))

means = np.array(means)
```

Nb. `means.append(value)` takes 'value' and adds it to the END of the list.

Plot the histogram of sample means

```
plt.hist(means, bins=30, edgecolor="black")
```

Questions

- What shape does this distribution have?
- What is the approximate mean of this distribution?

Part 4 — Central Limit Theorem (CLT) check

According to the CLT, for “large enough” n ,

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

Compute the empirical mean and variance of the sample means:

Questions

- Are the empirical values close to the theoretical ones?

Part 5 — Standardization

Define the standardized variable

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

Compute Z for each experiment:

```
z = (means - mu) / (sigma / np.sqrt(n))
```

Plot the histogram:

```
plt.hist(z, bins=30, edgecolor="black")
```